Research with AI Agents

What Works, What It Costs, and What an Institute Needs

Lukas Theissinger

October 30, 2026

Outline

OpenAI and Agent Usage

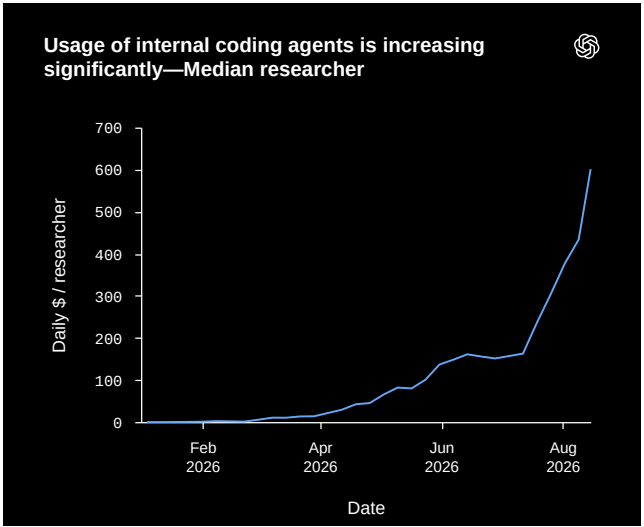
Owning Intelligence and Infrastructure at Lamarr

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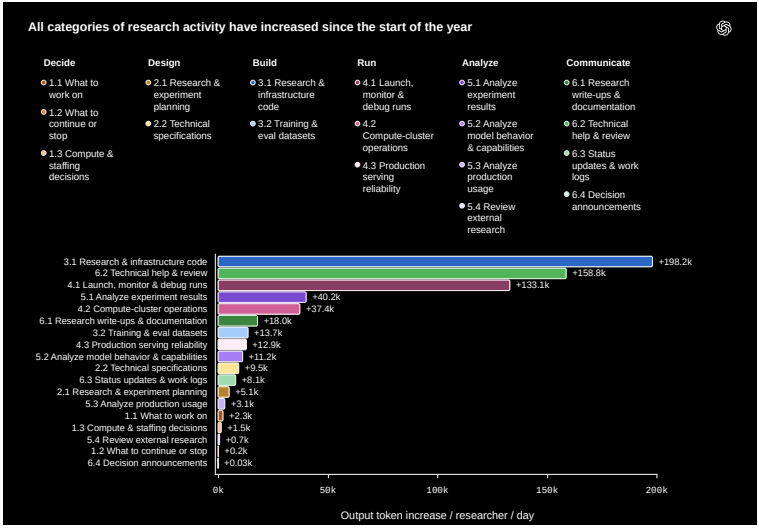
Owning Intelligence and Infrastructure at Lamarr

FOMO Headline: OpenAI Agent Usage



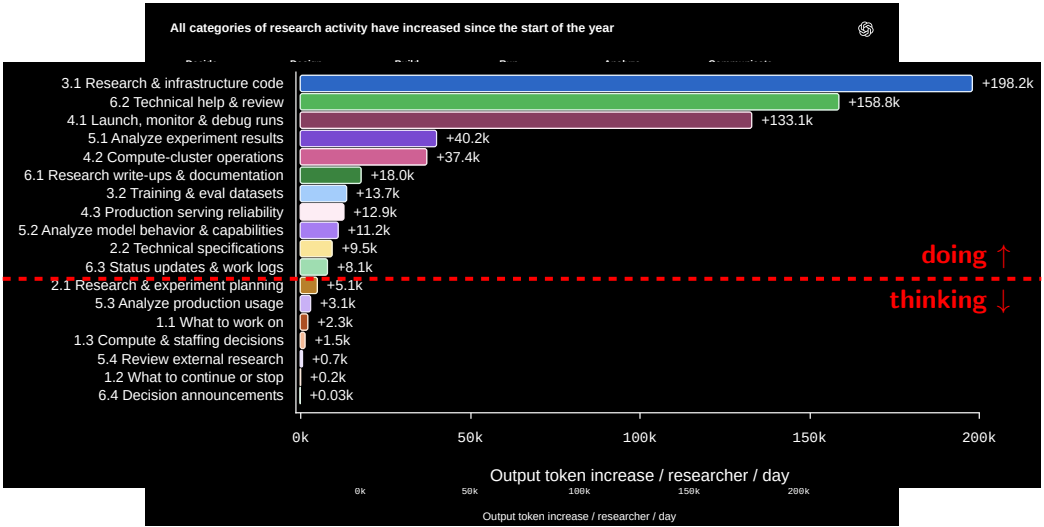
Median researcher, daily $ spent on internal coding agents. Source: OpenAI, "Research acceleration" (self-reported).

What do agents do at OpenAI?



Increase in output tokens per researcher per day, by research activity. Source: OpenAI, "Research acceleration" (self-reported).

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Tim Dettmers' hypotheses, read against the OpenAI usage data

What the OpenAI chart suggests

All 18 activities went up: agents are used everywhere, but how much and how exactly is still open.

Sources: timdettmers.com/2026/01/13/use-agents-or-be-left-behind; OpenAI, "Research acceleration" (self-reported).

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Field Report: Automating Exercises

1. **The problem.** PhD student, tutoring an exercise class every two weeks:
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4. **Ask your agent to try it:** `github.com/Newida/AutomatingExercises` **Avoid:** Don't do anything BY HAND on this project. If you want to change it, ASK YOUR AGENT to change it.

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Owning Intelligence and Infrastructure at Lamarr

Automating at Bigger Scale

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Collect receipts, fill in the university form, look up per-diem rules, fix what admin sends back: say **3 trips** $\times$ **3 h per person per year**. An agent that reads the receipts and files the form correctly needs $\sim$**200 h** (rules, forms, access to the admin system).

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institute, 10 000 people	$10\,000 \times 9\,\text{h} = 90\,000\,\text{h/year}$	200 h	**pays $>$400$\times$**

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Dear management: do not assume that employees will continue spending their own money. We are bottlenecking ourselves: we don't want to spend 100 € per month just to do our work, so we spend 20 €, effectively leaving performance on the table. Also, people will just switch to other institutes, even outside Germany, that supply them with the AI they want and need, for free, to get important work done.

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So let's talk about costs.

Decision 1: Subscription or API?

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Rule of thumb: people → subscription; pipelines and batch jobs → API.

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Small open models are clearly behind. The largest open models are much closer to the frontier; on our tasks they are **still untested**.

Self-Hosting for the Institute: Pros and Cons

Quality and performance

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Rough comparison

2 nodes $\times$ 300 k€ over 4 years $\approx$ 12.5 k€/month $\approx$ **300 people** $\times$ **40 €** $\approx$ **125 people** $\times$ **100 €**

With Lamarr's compute, a frontier-class open model for everyone is realistic: it pays off through privacy, scale and no caps; the closed models keep a lead of months.

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